

electrolyte A substance that conducts electricity when molten or in solution.

electromagnetic induction See *induction*, senses 2 and 3.

electromagnetic radiation Waves of energy that are in the form of electric and magnetic fields vibrating at right angles to each other.

electromagnetic rotation Mechanical rotation produced by electromagnetic means.

electromagnetic spectrum The complete range of electromagnetic radiation, including (from the highest to the lowest frequency and energy): gamma rays, X-rays, ultraviolet radiation, visible light, infrared radiation, microwaves, and radio waves.

electromagnetism The physics of the electromagnetic field created by the interaction of electricity and magnetism.

electromotive force (emf) The potential difference of a battery or generator, which "pushes" an electric current around a *circuit*.

electron A tiny subatomic particle with a negative electric charge. Electrons are leptons, with about a thousandth of the mass of a proton or neutron. They orbit the nuclei of atoms in a cloud, and their movement is responsible for electric currents in circuits.

electron micrograph Permanent magnified image of an object obtained using an electron microscope.

electron microscope A microscope that uses a beam of electrons instead of light to obtain a magnified image of an object.

electron shell One of the layers in which electrons orbit around the nucleus of an atom.

electron volt A small unit of energy, convenient when discussing the energies of subatomic particles.

electrophoresis A technique for analyzing and separating large molecules and small particles by using the different speeds at which they move through a medium when electricity is supplied.

electroscope An instrument demonstrating the presence of electric charge.

electrostatic Relating to stationary electric charges.

electrostatic field The field of force surrounding a stationary electrically charged object.

element In chemistry, a substance whose atoms are all of the same kind (that is, all have the same number of protons in their nuclei).

ellipse A flat symmetrical oval shape or outline, like a flattened circle.

embryo An early stage in the development of a new individual (animal or plant). In humans, an embryo more than eight weeks old is called a fetus.

emission lines Bright lines in the spectrum of light emitted by a body, usually indicating the presence of particular elements.

endangered species A species of living thing that is at risk of becoming extinct.

endocrinology The study of hormones and of the endocrine glands that produce them.

endoscope Any of various instruments for directly viewing inside the body.

energy Traditionally described as the capacity to do work, energy is difficult to

define fully but can be regarded as the agent that can produce change in the Universe.

entanglement In quantum physics, the linking of two particles as one, so that when the particles move apart, a change in one instantly causes a change in the other.

environment The surroundings of living things, sometimes including the living things themselves.

enzyme A catalyst in living things that increases the speed of a particular biochemical reaction. There are thousands of different kinds, nearly all of them proteins.

equinox Either of the twice-yearly moments when the Sun crosses the celestial equator and the day-length of the Northern and Southern Hemispheres is momentarily equal.

escapement Mechanism in a clock or watch that allows its motive power to be released in a way that provides exactly timed motion.

eukaryotic cell The typical cell of an animal or plant, in which the genes are contained in a nucleus. See also *prokaryotic cell*.

evolution The gradual process by which living things develop and change over a long period of time.

evolutionary biology The study of evolution and its related areas of biology.

exoplanet A planet that orbits a star other than the Sun; also known as an extrasolar planet.

exothermic Of a chemical reaction, resulting in the release of heat.

Fahrenheit scale The temperature scale named after Gabriel Fahrenheit, in which, under normal conditions, water freezes at 32° and boils at 212°.

faience Pottery decorated with glazes.

fermion Any of the group of subatomic particles associated with matter, such as electrons, quarks, and protons, rather than those that carry force, such as bosons.

Ferrel cell The circulation of the atmosphere in midlatitudes that brings westerly winds at surface level and returning easterly winds at high level.

fertilization The joining of two gametes as the first stage of producing a new organism.

fetus The unborn, developing offspring of a mammal. In humans, it covers pregnancy after the first eight weeks.

field of force A condition produced in the space around (for example) a magnet (magnetic field) or an electric charge (electric field), which can be diagrammed as curved lines showing the direction in which any nearby object influenced by these forces will tend to be moved.

filament See *anther*.

fissile A term referring to certain atomic nuclei, such as one type of uranium, which are capable of being split into two roughly equal parts when bombarded with neutrons.

flintlock mechanism A mechanism for discharging a gun by using the spark obtained from a flint striking metal.

fluid A substance that can flow, including solids, liquids, or plasmas.

FM Frequency modulation. The transmission of a signal by changing the frequency of the carrier wave, such as a radio wave.

formula (plural: **formulas** or **formulae**)

[1] In chemistry, a set of symbols that represent the makeup of a substance. [2] A set of mathematical symbols expressing a rule, principle, or method for finding an answer.

fossil The long-preserved remains of a living thing, especially when it has been mineralized (turned to stone).

Fraunhofer lines Dark lines in the spectrum of the Sun and other stars. They show where particular chemical elements in a star's outer layers are absorbing light from the stars.

freezing point The temperature at which a given liquid freezes. Freezing point also depends on pressure.

friction A force that resists or stops the movement of objects in contact with each other. The friction between an object and a fluid, such as air or water, is known as drag.

fundamental particle Also called an elementary particle, any subatomic particle such as electrons, which it is believed does not consist of simpler particles. Electrons are an example, but not protons or neutrons since they are made of quarks.

fuse [1] A safety device used in electrical circuits, such as a thin wire which melts if too much current passes through it. [2] A cord or other device that can be ignited or activated and used to set off an explosive.

Gaia hypothesis The concept that all the living things and physical components of Earth interact to form a complex self-regulating system, like a huge organism.

galaxy A huge grouping of stars, dust, and gas, all loosely held together by gravity. Our galaxy is called the Milky Way.

gamete A sex cell, such as a sperm or egg cell. Gametes have half the number of *chromosomes* of most other cells (see *haploid cell*) so that, when they join together in fertilization, the normal number of chromosomes is restored.

gametocytes Cells that represent an early stage in the production of gametes.

ganglion Concentration of nerve cell bodies, more so outside the central nervous system.

Geiger counter An instrument used to detect and measure radioactivity.

gene The basic unit of inheritance in living things, a segment of DNA (or RNA in some viruses) that typically codes for making a particular protein and also incorporates features allowing it to be switched on and off.

gene map A plot of the sequence of genes along an entire strand of DNA.

gene sequencing Finding out the order of *bases* in the DNA of particular genes.

generator A device that converts mechanical energy into electrical energy.

genetic code The code by which sequences of DNA "spell out" the recipe for making a particular protein. See also *codon*.

genetic drift Change in the overall genetic composition of a population as a result of random events rather than natural selection.

genetic engineering Techniques of artificially modifying an organism's characteristics by manipulating its genetic material.

genetic fingerprinting The analysis of a DNA sample to identify who it belongs to.

genome The complete set of genes for an organism.

genotype The genetic makeup of an organism.

geological period One of the time divisions (such as the Jurassic period) into which Earth's history is divided.

geostationary Term applied to a satellite orbiting Earth at the same rate as Earth rotates, so staying above one particular point on the surface.

geothermal Relating to the internal heat of Earth, or energy obtained from it.

germ theory The theory that infectious living agents (germs) cause many diseases.

glaciation Coverage of land areas by glaciers and ice caps.

gluons Particles within protons and neutrons that hold their component quarks together.

gravitational force The force of gravity. It is regarded as one of the four fundamental forces in the Universe.

gravitational lensing Phenomenon where the gravity of a large astronomical object can bend light coming from another behind it, sometimes resulting in several images of the more distant object being visible.

gravity The tendency of every body possessing mass to attract every other body.

greenhouse effect Heating effect due to heat rays from the ground being absorbed by some of the gases in the atmosphere.

greenhouse gases Gases that lead to the greenhouse effect, including water vapor, carbon dioxide, and methane.

habitat The environment where a particular living thing occurs naturally.

Hadley cell The circulation pattern of the atmosphere in the tropics, which brings north- and southeasterly trade winds toward the equator at surface level and returns westerly winds at high level.

half-life [1] The time taken for the radioactive emission of any particular radioactive material to drop to half of its initial value. [2] The time taken for a drug in the body to reduce to half of its original concentration.

haploid cell A cell containing only one copy of each chromosome.

hemoglobin An iron-containing protein that is the carrier of oxygen in the blood.

heredity The passing on of characteristics from one generation to the next.

hertz (Hz) The SI unit of frequency. One hertz is one cycle per second.

histology The study of tissues of the body.

homeobox A sequence of DNA that is part of the genes controlling body development in animals, plants, and other organisms.

hominid Any member of the primate family *Hominioidea*, including humans.

horology The science of clockmaking and of measuring time.

H-R diagram Hertzsprung–Russell diagram. A chart showing how stars evolve from one type to another over time.

html Hypertext markup language. The main computer language used on websites.